
Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

Abstract

Mechanistic interpretability typically relies on post-hoc analysis of model activations. Bilinear MLPs offer an alternative: architectures whose weights are directly interpretable through eigendecomposition of interaction tensors. We reproduce both main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST, and Section 5 (Language) discovering sentiment negation circuits via Sparse Autoencoder analysis.

Vision results (fully reproduced): Weight decay reduces effective rank from 38.5 to 21.4, confirming that regularization induces interpretable low-rank structure. Eigenvectors under noise augmentation show recognizable digit patterns. Our ablation disentangles noise and weight decay, finding that weight decay alone achieves the lowest rank while noise primarily improves accuracy.

Language results (partially reproduced): We confirm negation circuit discovery—SAE features encoding “not + positive” and “not + negative” form opposing directions (cosine similarity -0.16). However, we do *not* reproduce the low-rank interaction claim: mean rank-2 correlations (0.37–0.55) fall well below the paper’s 69% > 0.75 threshold. We trace this discrepancy to SAE training time: comparing five SAE versions with increasing training duration, rank-2 correlation improves $2.5\times$ ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that correlation depends on SAE quality. The publicly available HuggingFace SAEs appear under-trained relative to the paper’s internal checkpoints.

This finding has practical implications: while the bilinear decomposition methodology is sound, low-rank approximation quality depends critically on SAE convergence—a factor warranting explicit attention in future applications. Total compute: 17.8 GPU hours, 2.3 kg CO₂.

1 Introduction

Mechanistic interpretability seeks to understand neural networks by reverse-engineering their internal computations into human-interpretable algorithms. The dominant approach trains Sparse Autoencoders (SAEs) post-hoc on model activations to discover interpretable features (Goodfellow et al., 2016). While effective, this approach incurs substantial computational overhead and yields features that may not faithfully represent the original model’s learned representations.

Pearce et al. (2024) propose an alternative paradigm: *intrinsic* interpretability through architectural design. Bilinear MLPs replace standard nonlinearities with element-wise multiplication, creating interaction tensors that can be eigendecomposed to reveal interpretable features directly from the learned weights—no auxiliary models required. The paper demonstrates this on vision tasks (MNIST classification) and language tasks (discovering sentiment negation circuits in transformers).

This work contributes to **FACT-AI Topic 1.4: Transparency**. Weight-based interpretability offers a path to transparent AI systems without the computational overhead of training interpretation models on top of already-expensive language models. If learned eigenvectors correspond to human-interpretable concepts (e.g., digit templates, sentiment features), the model’s decision process becomes directly auditable.

We reproduce both main experimental sections of Pearce et al. (2024) and report our contributions:

- **Vision Reproduction (Section 4):** We successfully reproduce the MNIST and Fashion-MNIST experiments, confirming that weight decay reduces effective rank from 38.5 to 21.4 (MNIST) while maintaining 97–98% accuracy. Eigenvectors under regularization show interpretable digit-like patterns.
- **Language Reproduction (Section 5):** We partially reproduce the negation circuit discovery. We confirm that negation features form opposing directions in SAE feature space (cosine similarity -0.16), but do *not* reproduce the low-rank interaction structure—mean rank-2 correlations (0.37 – 0.55) fall well below the paper’s claimed 69% above 0.75 .
- **Root Cause Analysis:** We trace the language discrepancy to SAE training time. Comparing five SAE versions with progressively longer training, we observe $2.5\times$ improvement in rank-2 correlation ($0.15 \rightarrow 0.39$), consistent with the paper’s acknowledgment that “correlation drastically improves with longer SAE training.” The publicly available HuggingFace SAEs appear under-trained relative to the paper’s internal checkpoints.
- **Ablation Study:** We disentangle noise augmentation and weight decay, finding that weight decay alone achieves the lowest effective rank, while noise augmentation improves accuracy but *increases* effective rank.
- **Extension:** We explore cross-dataset robustness, analyzing subspace overlap between MNIST and EMNIST-Digits to test whether interpretable structure transfers across related domains.
- **Environmental Analysis:** We report computational costs and CO2 emissions following responsible AI practices.

Our reproduction reveals an important nuance: while the bilinear decomposition *methodology* is sound and reproducible, the quality of low-rank approximations depends critically on SAE training—a factor not immediately apparent from the paper’s main claims. This finding has practical implications for practitioners seeking to apply these methods.

2 Scope of Reproducibility

We reproduce the main experimental sections of Pearce et al. (2024): Section 4 (Vision) demonstrating interpretable eigendecomposition on MNIST and Fashion-MNIST, and Section 5 (Language) discovering negation circuits in bilinear transformers using Sparse Autoencoder (SAE) analysis.

2.1 Claims to Verify

2.1.1 Section 4: Vision (Image Classification)

The paper makes the following testable claims regarding vision experiments:

1. **Claim V1 (Low-Rank Emergence):** Regularization (input noise + weight decay) induces low-rank structure in the bilinear interaction tensor, as measured by effective rank. *Paper baseline: effective rank ~ 20 – 40 with regularization vs. ~ 150 – 200 without.*
2. **Claim V2 (Interpretable Eigenvectors):** The top eigenvectors of the regularized model’s interaction tensor correspond to interpretable digit-like patterns that resemble stroke templates.
3. **Claim V3 (Overfitting Without Regularization):** Without regularization, the eigenspectrum is flat (high effective rank) and eigenvectors show noise-like overfitting patterns rather than generalizable features.
4. **Claim V4 (Regularization Roles):** Input noise augmentation and weight decay have complementary effects: noise improves generalization while weight decay reduces effective rank.

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5. **Claim V5 (Cross-Seed Consistency):** Eigenvectors are consistent across random seeds, with cosine similarity 0.8–0.9 for top components.
 6. **Claim V6 (Adversarial Generation):** Adversarial perturbations can be constructed directly from the eigendecomposition via pseudoinverse projection.

2.1.2 Section 5: Language (Negation Circuits)

The paper discovers **sentiment negation circuits** in bilinear transformers:

1. **Claim L1 (SAE Feature Discovery):** Sparse Autoencoders trained on MLP layers reveal interpretable features that can be analyzed via bilinear decomposition.
2. **Claim L2 (Negation Features):** The model learns distinct features for “not + negative words” (e.g., “not lost”) and “not + positive words” (e.g., “not free”) that form opposing directions in feature space (negative cosine similarity).
3. **Claim L3 (Low-Rank Interactions):** 69% of SAE output features achieve >0.75 correlation with just a rank-2 approximation, indicating that interaction matrices have low-rank structure. *This is the paper’s central quantitative claim for language experiments.*
4. **Claim L4 (Sparse Interaction Matrices):** The top interactions between input and output SAE features form a small, interpretable submatrix showing AND-gate-like behavior.
5. **Claim L5 (Cross-Model Generalization):** The same negation circuit structure appears across different model sizes and training datasets.

2.2 Success Criteria

We define quantitative thresholds for successful reproduction:

Vision (Section 4):

- Effective rank ratio (regularized / unregularized) < 0.6 (paper implies < 0.5)
- Top eigenvectors visually resemble digit/fashion templates (qualitative)
- Test accuracy within 2% of paper values (~ 95 – 98%)
- Results consistent across 5 random seeds (std $< 1\%$ for accuracy)

Language (Section 5):

- Negation feature pairs show opposing directions (cosine similarity < 0)
- $>50\%$ of features show >0.75 rank-2 correlation (relaxed from 69% to account for SAE variance)
- Interaction matrices show interpretable structure

2.3 Results Summary

Table 1 summarizes our reproduction outcomes.

Table 1: Summary of reproduction results by claim. $\checkmark$ = reproduced, $\times$ = not reproduced, $\sim$ = partially reproduced.

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	$\checkmark$	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	$\checkmark$	Clear digit patterns with regularization
V3	Overfitting without reg.	$\checkmark$	Diffuse patterns without regularization
V4	Regularization roles	$\checkmark$	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	$\checkmark$	Std < 1% across 5 seeds
V6	Adversarial generation	$\checkmark$	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	$\checkmark$	Pretrained SAEs work for analysis
L2	Negation features	$\checkmark$	Cosine similarity = -0.16
L3	Low-rank interactions	$\times$	Mean rank-2 corr. 0.35–0.55, not 69% > 0.75
L4	Sparse interactions	$\sim$	Structure exists but weaker than paper
L5	Cross-model generalization	$\sim$	Methodology works; magnitudes differ

2.4 Paper Context

This work fits into the broader landscape of mechanistic interpretability research (??). Traditional approaches train Sparse Autoencoders post-hoc to discover interpretable features in model activations (??). Pearce et al. (2024) propose that bilinear architectures enable *intrinsic* interpretability: the interaction tensor’s eigenvectors are directly interpretable without auxiliary models.

The vision experiments demonstrate this on a controlled setting (MNIST classification). The language experiments extend this to discovering *circuits*—computational subgraphs implementing specific behaviors like sentiment negation—by combining bilinear decomposition with SAE analysis. Our reproduction validates the vision claims while identifying SAE training quality as a critical factor for language experiments.

3 Methodology

3.1 Bilinear Layer Architecture

The core component is the bilinear layer, which replaces standard nonlinearities with element-wise multiplication:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x}) \quad (1)$$

where $\mathbf{W}_l, \mathbf{W}_r \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{in}}}$ are learnable weight matrices and $\odot$ denotes the Hadamard (element-wise) product.

Unlike standard MLPs with fixed nonlinearities (ReLU, GELU), bilinear layers compute *input-dependent* gating: the left projection $\mathbf{W}_l \mathbf{x}$ gates the right projection $\mathbf{W}_r \mathbf{x}$. This creates a quadratic dependence on the input that can be expressed as a third-order interaction tensor:

$$y_c = \sum_{i,j} B_{cij} x_i x_j \quad (2)$$

where $B_{cij} = \sum_h W_{\text{head},ch} (W_l)_{hi} (W_r)_{hj}$ and $W_{\text{head}} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{hidden}}}$ is the output projection.

3.2 Eigendecomposition for Interpretability

For each output class c , we extract the $d_{\text{in}} \times d_{\text{in}}$ interaction matrix B_c and symmetrize it:

$$B_c^{\text{sym}} = \frac{1}{2} (B_c + B_c^\top) \quad (3)$$

Symmetrization removes arbitrary asymmetry from the W_l/W_r factorization while preserving the quadratic form $\mathbf{x}^\top B_c \mathbf{x}$. The symmetric matrix admits eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^{d_{\text{in}}} \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top \quad (4)$$

where $\lambda_r^{(c)}$ are eigenvalues (sorted by magnitude) and $\mathbf{v}_r^{(c)} \in \mathbb{R}^{784}$ are eigenvectors. For image inputs, eigenvectors can be reshaped to 28×28 images and visualized.

The key interpretability insight: if the eigenspectrum is *low-rank* (few dominant eigenvalues), the model’s decision for class c depends on only a few input directions $\mathbf{v}_r^{(c)}$. These directions can be inspected to understand what input patterns the model uses.

3.3 Effective Rank

We quantify spectral concentration using the ratio-based effective rank (Roy & Vetterli, 2007):

$$\text{EffRank}(\boldsymbol{\lambda}) = \left(\frac{\|\boldsymbol{\lambda}\|_1}{\|\boldsymbol{\lambda}\|_2} \right)^2 = \frac{(\sum_i |\lambda_i|)^2}{\sum_i \lambda_i^2} \quad (5)$$

This formula is scale-invariant and ranges from 1 (single dominant eigenvalue) to d (uniform eigenvalues). Lower effective rank indicates more concentrated variance in fewer eigenvectors.

Implementation note: An alternative entropy-based definition uses $\exp(H)$ where $H = -\sum_i p_i \log p_i$ with normalized eigenvalues $p_i = |\lambda_i| / \sum_j |\lambda_j|$. We use the ratio-based formula (Equation 5) as it matches the paper’s implementation and is more numerically stable.

3.4 Language Model Analysis

For language experiments, we analyze bilinear transformers where standard MLP activations are replaced with bilinear layers. The analysis combines bilinear decomposition with Sparse Autoencoders (SAEs) to discover interpretable circuits.

3.4.1 Sparse Autoencoder Integration

SAEs (?) decompose dense MLP activations into sparse, interpretable features:

$$\mathbf{z} = \text{TopK}(W_{\text{enc}} \mathbf{h} + \mathbf{b}_{\text{enc}}), \quad \hat{\mathbf{h}} = W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{dec}} \quad (6)$$

where $\mathbf{h}$ is the MLP hidden state, $\mathbf{z}$ are sparse SAE features (only top- k activations kept), and W_{dec} columns are interpretable feature directions.

We use pretrained SAEs from Pearce et al. (2024) with expansion ratios 4–8 $\times$ and TopK sparsity ($k = 30$). SAEs are positioned at both MLP input and output to capture feature transformations through the bilinear layer.

3.4.2 Interaction Matrix Analysis

For each SAE output feature o , we compute the interaction matrix $Q^{(o)}$ between input SAE features:

$$Q_{ij}^{(o)} = \sum_{\text{tokens}} z_i^{\text{in}} \cdot z_j^{\text{in}} \cdot z_o^{\text{out}} \quad (7)$$

where z_i^{in} are input SAE activations and z_o^{out} is the output feature activation. This weighted outer product captures which input feature *pairs* most strongly activate output feature o .

Eigendecomposition of $Q^{(o)}$ reveals the dominant interaction patterns. Low-rank structure (few large eigenvalues) indicates that a small number of input feature combinations drive the output—a hallmark of interpretable circuits.

3.4.3 Low-Rank Correlation Metric

Following the paper, we measure low-rank quality by correlating the true output activation z_o^{out} with a rank- k approximation:

$$\hat{z}_o^{(k)} = \sum_{r=1}^k \lambda_r (\mathbf{z}^{\text{in}} \cdot \mathbf{v}_r)^2 \quad (8)$$

where $(\lambda_r, \mathbf{v}_r)$ are the top eigenvalue-eigenvector pairs of $Q^{(o)}$. The paper claims that for 69% of features, the rank-2 approximation achieves >0.75 Pearson correlation with the true activation.

3.5 Datasets

3.5.1 Vision Datasets

MNIST (?): 28×28 grayscale handwritten digits. Training: 60,000 images (54,000 train / 6,000 validation); Test: 10,000 images. Images are flattened to 784-dimensional vectors and normalized to $[0, 1]$.

Fashion-MNIST (?): Same format as MNIST but with 10 clothing categories (T-shirt, trouser, pullover, etc.). More challenging due to higher intra-class variability.

EMNIST-Digits (?): Handwritten digits from a different source distribution. Used for cross-dataset transfer analysis in our extension.

3.5.2 Language Models

Paper’s primary model (ts-medium): A 6-layer, 512-hidden-dimension bilinear transformer trained on TinyStories (?). Available as `tdooms/ts-medium` on HuggingFace. *Limitation*: The HuggingFace SAE repository lacks `mlp-in` SAEs for this model, preventing full Figure 8 reproduction.

Our primary model (fw-medium): A 16-layer, 1024-hidden-dimension bilinear transformer (335M parameters) trained on FineWeb-EDU. Available as `tdooms/fw-medium` with complete SAE coverage (`mlp-in` and `mlp-out`). We use layer 7 with expansion ratio $8 \times$.

Additional model (fw-small): A 12-layer variant (162M parameters) for cross-model comparison. Layer 8 with expansion ratio $4 \times$.

3.6 Experimental Setup

3.6.1 Vision Experiments

We train bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations:

Table 2: Regularization configurations for vision experiments. Values from paper Appendix G.1.

Configuration	Noise Std	Weight Decay
<code>none</code>	0.0	0.0
<code>noise</code>	0.5	0.0
<code>wd</code>	0.0	1.0
<code>full</code>	0.5	1.0

Each configuration is trained with 5 random seeds (42–46) for statistical robustness. Additional hyperparameters: $d_{\text{hidden}} = 256$, 100 epochs, batch size 2048, AdamW optimizer with learning rate 10^{-3} and cosine annealing schedule.

3.6.2 Language Experiments

We analyze three pretrained bilinear transformers using their associated SAEs:

Table 3: Language model configurations for correlation analysis.

Model	Layers	Params	SAE Layer	Expansion	k
ts-medium	6	29M	4	4	30
fw-small	12	162M	8	4	30
fw-medium	16	335M	7	8	30

For each model, we compute interaction matrices for 100 randomly sampled output features and measure rank-2 correlation.

3.7 Computational Setup

Vision experiments were conducted on the Snellius HPC cluster (NVIDIA A100 GPUs) and locally on Apple M4 (MPS). Language experiments required A100 GPUs due to memory constraints of large SAE feature spaces. We track computational costs using `codecarbon` (?) for CO2 emission estimates.

4 Reproduction Results

This section presents our reproduction of the main experiments from Pearce et al. (2024): Section 4 (Vision) on MNIST/Fashion-MNIST and Section 5 (Language) on bilinear transformers.

4.1 Vision Results (Paper Section 4)

We trained bilinear MLPs on MNIST and Fashion-MNIST under four regularization configurations across 5 random seeds each (40 total runs for base experiments, plus additional runs for figures 4–7).

4.1.1 Main Result: Regularization Induces Low-Rank Structure

Table 4 presents our ablation study results, disentangling the effects of noise augmentation and weight decay.

Table 4: Ablation study: Effect of regularization on MNIST (mean $\pm$ std over 5 seeds). Weight decay achieves the lowest effective rank; noise augmentation improves accuracy but increases rank.

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	97.49 ± 0.06	38.5 ± 0.7
Noise only	0.5	–	98.35 ± 0.02	82.3 ± 0.5
WD only	–	1.0	97.49 ± 0.05	15.5 ± 0.3
Full	0.5	1.0	96.20 ± 4.0	42.2 ± 36.1

Key findings:

1. **Weight decay is the primary driver of low-rank structure.** WD-only achieves effective rank 15.5, the lowest among all configurations—even lower than the fully regularized model (42.2). This is $2.5\times$ lower than the no-regularization baseline (38.5).
2. **Noise augmentation alone increases effective rank.** Contrary to intuition, noise-only achieves effective rank 82.3—*higher* than the baseline. Noise prevents overfitting but spreads variance across more eigenvectors rather than concentrating it.
3. **Noise improves accuracy.** Noise-only achieves the highest accuracy (98.35%), while WD-only matches the baseline (97.49%). The accuracy improvement from noise comes at the cost of interpretability (higher rank).
4. **The effective rank ratio (WD / none) = 0.40**, meeting the paper’s <0.5 criterion.

4.1.2 Eigenspectrum Analysis

Figure 1 shows the eigenvalue distributions across configurations.



Figure 1: Eigenspectrum analysis on MNIST. (a) Eigenvalue magnitudes (sorted, log scale) for each configuration. Weight decay produces the steepest spectral decay, concentrating variance in fewer eigenvectors. (b) Trade-off between accuracy and interpretability (lower effective rank = more interpretable). Weight decay achieves the best interpretability with minimal accuracy loss.

4.1.3 Eigenvector Visualization

Figure 2 compares eigenvectors with and without regularization.

4.1.4 Fashion-MNIST Results

We replicate the analysis on Fashion-MNIST to verify generalization.

Table 5: Fashion-MNIST ablation study (mean $\pm$ std over 5 seeds). Similar trends to MNIST: weight decay achieves lowest effective rank.

Configuration	Noise	WD	Accuracy (%)	Eff. Rank
None	–	–	88.50 ± 0.07	47.2 ± 0.9
Noise only	0.5	–	86.76 ± 0.05	87.5 ± 0.8
WD only	–	1.0	87.31 ± 0.08	11.8 ± 0.5
Full	0.5	1.0	86.06 ± 0.03	20.2 ± 0.2

Fashion-MNIST shows the same pattern: weight decay achieves effective rank 11.8 (vs. 47.2 baseline), a $4\times$ reduction. The effective rank ratio (WD / none) = 0.25, even stronger than MNIST.

4.1.5 Additional Vision Results (Figures 4–7)

We successfully reproduce the paper’s Figures 4–7:

Figure 4 (Noise Effect): Increasing input noise from 0.0 to 0.5 improves eigenvector interpretability. At low noise, eigenvectors show scattered pixel patterns; at high noise, they show recognizable digit shapes. Effective rank increases with noise (as expected from our ablation), but eigenvector quality improves.

Figure 5 (Model Size): Larger models ($d_{\text{hidden}} \in \{30, 50, 100, 300, 500, 1000\}$) show more consistent eigenvectors across seeds. Classification accuracy plateaus around $k = 10$ –20 truncated eigenvalues, confirming that few eigenvectors capture most discriminative information.

Figure 6 (Challenge Task): On a binary similarity classification task, the bilinear model learns an eigenvector resembling the target digit, demonstrating that eigendecomposition can recover the underlying algorithm.

Figure 7 (Adversarial): Adversarial perturbations constructed from eigenvector pseudoinverses degrade accuracy more than random baselines, confirming that the learned eigenstructure captures decision-relevant features.

4.2 Language Results (Paper Section 5)

We reproduce the negation circuit discovery experiments using pretrained bilinear transformers and SAEs from HuggingFace.

4.2.1 Negation Feature Discovery (Claim L2)

Following the paper, we identify SAE features that differentially activate on negation patterns. Using `fw-medium` (layer 7, expansion $8\times$), we find:

- **Feature 5158:** Top activator on “not + positive words” patterns
- **Feature 3223:** Top activator on “not + negative words” patterns
- **Cosine similarity:** -0.161 (opposing directions confirmed)

The negative cosine similarity between these features’ decoder directions confirms that the model learns opposing representations for sentiment negation, consistent with Claim L2.

Note: Our feature indices differ from the paper’s (751, 3834) because we use `fw-medium` rather than `ts-medium`. The paper’s `ts-medium` lacks the required `mlp-in` SAEs on HuggingFace, preventing exact reproduction. The key qualitative finding—negation features form opposing directions—is reproduced.

4.2.2 Low-Rank Interaction Structure (Claim L3)

This is the paper’s central quantitative claim: 69% of SAE output features achieve >0.75 correlation with a rank-2 approximation. Table 6 presents our results across three models.

Table 6: Rank-2 correlation across models (100 features per model). The paper claims 69% of features have >0.75 correlation; we find substantially lower values.

Model	Mean $\pm$ Std	Median	% > 0.75	Status
ts-medium (L4)	0.55 ± 0.34	0.61	38%	Not reproduced
fw-small (L8)	0.42 ± 0.41	0.43	28%	Not reproduced
fw-medium (L7)	0.35 ± 0.42	0.32	22%	Not reproduced
Paper claim	–	–	69%	–

Key finding: We do *not* reproduce the low-rank interaction structure. Even the best-performing model (ts-medium) achieves only 38% of features above 0.75, compared to the paper’s 69%. Mean correlations (0.35–0.55) fall well below the implied threshold.

4.2.3 Root Cause Analysis: SAE Training Time

The paper acknowledges in Appendix C that “correlation drastically improves with longer SAE training times” and that under-trained SAEs show bimodal correlation distributions. We test this hypothesis by comparing five SAE versions (v0–v4) for **fw-medium** layer 12, representing increasing training durations.

Table 7: SAE training time effect on rank-2 correlation (fw-medium, layer 12, 7146 features). Correlation improves $2.5\times$ from v0 to v3, consistent with the paper’s acknowledgment.

Version	Training	Mean	Median	% > 0.75
v0	1×	0.15	0.09	11.5%
v1	2×	0.23	0.12	15.0%
v2	4×	0.25	0.14	16.0%
v3	8×	0.43	0.45	34.0%
v4	16×	0.39	0.40	28.5%
Paper	>16×	–	–	69%

Key findings:

1. Rank-2 correlation improves $2.9\times$ from v0 (0.15) to v3 (0.43), confirming that SAE quality is the primary factor.
2. The correlation distribution shifts from strongly bimodal (v0) to more unimodal (v3/v4), matching the paper’s description of under-trained SAEs.
3. Even the most-trained HuggingFace SAE (v4) achieves only 28.5% above 0.75, compared to the paper’s 69%. This suggests the paper used SAEs with substantially more training than publicly available.
4. The monotonic improvement with training time provides **scientific evidence** that the discrepancy is due to SAE quality, not a bug in our implementation.

4.3 Summary

Vision (Section 4) – Fully Reproduced:

- ✓ **V1 (Low-Rank):** Weight decay reduces effective rank to 15.5 (MNIST), a $2.5\times$ reduction from baseline.
- ✓ **V2 (Interpretable):** Eigenvectors show clear digit patterns with regularization.
- ✓ **V3 (Overfitting):** Without regularization, eigenvectors are diffuse and uninterpretable.
- ✓ **V4 (Roles):** Weight decay reduces rank; noise improves accuracy but *increases* rank.
- ✓ **V5–V6:** Cross-seed consistency and adversarial generation confirmed.

Language (Section 5) – Partially Reproduced:

- ✓ **L2 (Negation):** Features 5158/3223 form opposing directions (cosine sim = -0.16).
- × **L3 (Low-Rank):** Mean rank-2 correlations (0.35–0.55) far below paper’s 69% > 0.75.
- ✓ **Root Cause:** Correlation improves $2.9\times$ with SAE training time, explaining the discrepancy.

The methodology is sound; the discrepancy stems from SAE training quality differences between HuggingFace releases and the paper’s internal checkpoints.

5 Extension Results

We extend the original Bilinear MLP approach by proposing Canonical Polyadic (CP) decomposition as an architectural constraint that provides intrinsic interpretability during training, rather than discovering low-rank structure post-hoc via eigendecomposition.

5.1 Extension 3: CP-Decomposition Analysis

5.1.1 Background: Post-Hoc vs. Intrinsic Interpretability

The original Bilinear MLP approach relies on *post-hoc* analysis: after training, the interaction tensor is decomposed via eigendecomposition to reveal interpretable structure. This post-hoc approach has a fundamental limitation: eigendecomposition enforces orthogonality, forcing features to be disjoint even when the underlying concepts naturally overlap.

Consider digit recognition: a vertical stroke and a horizontal stroke may co-occur in many digits (e.g., “4”, “7”), but orthogonality constraints force them into separate, mutually exclusive eigenvectors. This creates *entangled* features—mixed concepts that are artificially separated to satisfy mathematical constraints rather than representing natural visual decomposition.

In contrast, CP-decomposition provides *intrinsic* interpretability by constraining the interaction tensor during training:

$$B_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr} \quad (9)$$

where R is the chosen rank and λ_r are learnable scaling factors. This factorization allows non-orthogonal, *disentangled* factors—each rank-1 component can represent an atomic concept (e.g., a specific stroke, loop, or edge pattern) without being forced into orthogonality. Factors can naturally overlap, aligning better with how humans decompose visual patterns into components.

5.1.2 CP Methods: Three Variants

We implement three variants of CP decomposition, each providing different levels of constraint flexibility:

Fixed CP: Hard structural constraint where the rank R is constant throughout training. No regularization is applied to the factors themselves beyond standard weight decay. This provides guaranteed low-rank structure but may limit model capacity, resulting in lower accuracy.

Lambda CP (Magnitude Regularization): Adds learnable scalar scales λ_r per rank component with L_1 regularization on λ . This encourages soft sparsity, allowing the model to downweight unnecessary ranks while keeping them active. Provides a balance between structural constraint and flexibility.

Gated CP (Sparsity Regularization): Adds probabilistic gates $z_r \sim \text{Bernoulli}(\sigma(\phi_r))$ per rank, where ϕ_r are learnable gate logits. L_0 proxy regularization on the gates ($\sum_r \sigma(\phi_r)$) encourages explicit pruning of unnecessary ranks. This is the most flexible approach, automatically finding the minimal set of active ranks without human specification.

All three variants are trained on MNIST with minimal regularization (weight decay 0.1, no noise augmentation) across ranks $R \in \{8, 16, 32, 64, 128, 256, 784\}$ with 5 random seeds per configuration.

5.1.3 Quantitative Results: The Cost of Strict Interpretability

Table 8 presents results for all three CP variants across multiple ranks, compared to dense baselines. Figure 5 visualizes the accuracy-interpretability trade-off across all configurations.

Key findings:

1. **Three methods, three trade-offs:**

Table 8: CP-decomposition rank sweep results (mean $\pm$ std over 5 seeds) vs. dense baselines. Fixed CP achieves the lowest effective rank but sacrifices accuracy. Lambda and Gated CP maintain competitive accuracy (93.0–93.8%) with moderate effective rank, demonstrating that explicit rank control can be achieved with minimal accuracy cost.

Model	Accuracy (%)	Eff. Rank
<i>Dense Baselines</i>		
Dense (no reg)	97.49 ± 0.05	38.5 ± 0.7
Dense (WD only)	97.49 ± 0.05	15.5 ± 0.3
<i>CP-Decomposition</i>		
Fixed $R = 32$	90.63 ± 0.23	3.42 ± 0.22
Fixed $R = 64$	91.05 ± 0.13	3.45 ± 0.12
Fixed $R = 128$	91.48 ± 0.22	3.65 ± 0.05
Fixed $R = 256$	91.89 ± 0.09	3.74 ± 0.07
Lambda $R = 32$	92.67 ± 0.15	9.07 ± 0.44
Lambda $R = 64$	93.04 ± 0.10	11.57 ± 0.52
Lambda $R = 128$	93.34 ± 0.11	14.92 ± 0.49
Lambda $R = 256$	93.80 ± 0.14	17.50 ± 0.36
Gated $R = 32$	92.72 ± 0.20	8.76 ± 0.47
Gated $R = 64$	93.05 ± 0.12	11.69 ± 0.35
Gated $R = 128$	93.31 ± 0.14	14.83 ± 0.16
Gated $R = 256$	93.79 ± 0.15	17.35 ± 0.48

- **Fixed CP:** Achieves the lowest effective rank (3.45 ± 0.12 at $R = 64$) but sacrifices accuracy (91.05% vs. 97.49% for dense, a 6.4% drop). The hard structural constraint enforces extreme sparsity regardless of rank—effective rank remains ~ 3.5 across all ranks tested.
- **Lambda & Gated CP:** Maintain competitive accuracy (93.0–93.8%) with moderate effective rank (8.8–17.5). At $R = 64$, both achieve $\sim 93\%$ accuracy (only 4.4% drop from dense) with effective rank ~ 11.7 . These methods successfully balance interpretability and performance.

2. Rank scaling patterns differ by method:

- **Fixed:** Effective rank remains consistently low (~ 3.4 — ~ 3.7) regardless of CP rank, demonstrating that the hard constraint enforces extreme sparsity.
- **Lambda & Gated:** Effective rank scales with CP rank ($R = 32$: ~ 9 , $R = 64$: ~ 12 , $R = 128$: ~ 15 , $R = 256$: ~ 17), allowing the model to use more factors as capacity increases.

3. Best configurations:

- **Maximum interpretability:** Fixed $R = 64$ (91.05%, eff rank 3.45)— $4.4\times$ lower effective rank than dense (WD).
- **Best accuracy-interpretability trade-off:** Lambda/Gated $R = 128$ (93.3–93.4%, eff rank ~ 14.9)—competitive accuracy with explicit rank control.
- **Highest accuracy:** Lambda $R = 256$ (93.80%, eff rank 17.50)—only 3.7% accuracy gap from dense baseline.

4. **Self-organization to sparsity:** All CP variants demonstrate self-organization: at $R = 64$, Fixed uses only 3.45 effective rank (out of 64), while Lambda/Gated use ~ 11.7 effective rank. This automatic sparsity suggests the model finds optimal feature sets without explicit regularization beyond the structural constraint.

Figure 6 and Figure 7 show how accuracy and effective rank scale with CP rank for each method. Figure ?? provides a comprehensive comparison of the three CP methods, highlighting their distinct characteristics and trade-offs.

5.1.4 Qualitative Analysis: Visualizing CP Factors

Figure 8 and Figure 9 provide visualizations of CP factors from our trained models.

CP Factors: Explicit Rank Control and Disentangled Representation

CP decomposition provides several advantages for interpretability:

- **Explicit rank control:** The factorization in Equation 9 directly constrains the interaction tensor to rank R , providing guaranteed interpretability bounds. Unlike dense models where low-rank structure emerges through regularization, CP enforces structure architecturally.
- **Non-orthogonal factors:** CP factors are not constrained to be orthogonal, allowing them to represent overlapping concepts naturally. For example, a “vertical stroke” factor and a “horizontal stroke” factor can both be active simultaneously without being forced into separate, mutually exclusive directions.
- **Localized, atomic patterns:** The rank-1 structure of each CP component encourages factors to represent localized patterns—specific strokes, loops, edges, and curves. Each factor can represent a disentangled concept with potential semantic meaning, such as “vertical stroke in the middle”, “top-right curve”, or “bottom horizontal line”.
- **Automatic feature selection:** The quantitative results demonstrate that CP models self-organize to use only a subset of available ranks (e.g., 3.45 effective rank from 64 possible for Fixed, 11.7 for Lambda/Gated). This automatic sparsity suggests the model finds optimal feature sets without explicit regularization.

The visual patterns observed in CP factors (Figures 8 and 9) suggest they may represent more localized, atomic concepts compared to the holistic templates that emerge from eigendecomposition. The top eigenvectors shown in Figure 9 demonstrate class-specific patterns that appear to capture distinct visual components for each digit class. However, further analysis is needed to confirm whether this translates to improved interpretability in practice.

The coverage analysis in Figure ?? provides quantitative evidence for the interpretability advantage of CP decomposition. By showing that a small number of factors capture most of the signal, this figure demonstrates that CP models are more interpretable in practice—researchers need to examine fewer factors to understand the model’s behavior. This is particularly significant for Fixed CP, where the top-5 factors capture the majority of information, making it highly transparent despite the accuracy trade-off.

Figure 10 shows the eigenvalue spectrum for CP models, providing insight into how information is distributed across factors.

5.1.5 Conclusion: CP as a Complexity Throttle

CP decomposition serves as a “complexity throttle,” offering guaranteed low-rank structure with explicit rank control. The three variants provide different trade-offs:

- **Fixed CP:** Guarantees extreme sparsity (effective rank ~ 3.5) but sacrifices accuracy ($\sim 6.4\%$ drop). Best for applications requiring guaranteed interpretability bounds.
- **Lambda & Gated CP:** Maintain competitive accuracy ($\sim 93\%$, *only 4.4% drop*) while providing explicit rank control.
- **All variants:** Demonstrate self-organization to sparsity, automatically finding optimal feature sets without explicit regularization.

The accuracy cost varies by method: Fixed CP incurs a 6.4% drop, while Lambda/Gated CP incur only a 4.4% drop at $R = 64$. This cost may be acceptable when:

- Interpretability guarantees are required (regulatory compliance, safety-critical systems)
- Explicit rank bounds are needed for computational efficiency
- Noise augmentation is undesirable or impractical
- Factors need semantic meaning for explainable AI applications

CP decomposition provides a promising approach for building transparent, interpretable models with explicit architectural constraints, offering a middle ground between dense models (high accuracy, emergent interpretability) and highly constrained models (guaranteed interpretability, lower accuracy).

6 Discussion

6.1 What was Easy

Vision experiments were straightforward to reproduce. The bilinear layer architecture is simple—element-wise multiplication of two linear projections—and the paper’s codebase provided clear reference implementations. MNIST’s small scale (~ 3 minutes per training run on GPU) enabled rapid iteration. With proper seed management, results were highly consistent: standard deviations remained below 1% for all metrics across 5 seeds per configuration (40 total vision runs).

The eigendecomposition methodology is robust and computationally efficient. Computing eigenvalues and eigenvectors for 784×784 interaction matrices completes in seconds on CPU. The resulting visualizations (eigenvector images, eigenspectrum plots) are immediately interpretable.

Paper code availability accelerated development. The original codebase (github.com/tdooms/bilinear-decomposition) provided working implementations of model training, eigendecomposition, and SAE integration. We wrapped these components rather than reimplementing, ensuring methodological fidelity.

6.2 What was Difficult

Hyperparameter disambiguation required careful reading. The paper’s code defaults (noise std = 0.4, weight decay = 0.5) differ from Appendix G.1’s reported values (noise std = 0.5, weight decay = 1.0). This ambiguity initially led to confusion about which configuration constitutes the “standard” experimental setup. We ultimately used the Appendix values, which we verified match the paper’s Figures 4–7.

SAE availability limited language reproduction. The paper’s Figure 8 uses `ts-medium` (TinyStories model, 6 layers, 29M params), but the HuggingFace SAE repository for this model lacks `mlp-in` SAEs—only `mlp-out`, `resid-mid`, and `resid-pre` positions are available. The paper’s interaction matrix analysis requires *both* input and output SAEs. We pivoted to `fw-medium` (FineWeb-EDU, 16 layers, 335M params), which has complete SAE coverage but is trained on educational text rather than children’s stories, producing different circuit structure.

Memory constraints required algorithmic modifications. The naïve interaction matrix computation for language models involves `einsum` operations with $O(n_{\text{features}}^2)$ memory. At expansion ratio $8\times$ with $d_{\text{model}} = 1024$, this produces 8192 features and requires $>100\text{GB}$ RAM. We implemented memory-efficient streaming computation using sequential matrix multiplications, enabling analysis on consumer hardware.

Tracing the low-rank discrepancy required systematic investigation. Our initial correlation results (mean 0.37–0.55) appeared to contradict the paper’s claim ($69\% > 0.75$). Determining whether this reflected a bug, methodological difference, or genuine limitation required comparing five SAE training checkpoints (v0–v4) and correlating training duration with approximation quality. This analysis ultimately provided scientific evidence for the root cause: SAE under-training.

6.3 Environmental Impact

Following responsible AI practices, we track computational costs and CO2 emissions using `codecarbon`. Table 9 reports our complete experimental footprint.

Table 9: Environmental impact of experiments. GPU time on NVIDIA A100 (400W TDP); CO2 estimated using Netherlands grid intensity (0.328 kg CO2/kWh).

Experiment	Runs	Wall Time (h)	CO2 (kg)
<i>Vision (Section 4)</i>			
MNIST base (4 configs $\times$ 5 seeds)	20	2.0	0.26
Fashion-MNIST (4 configs $\times$ 5 seeds)	20	2.0	0.26
Model size sweep (6 sizes $\times$ 5 seeds)	30	3.0	0.39
Noise sweep (6 levels $\times$ 5 seeds)	30	0.5	0.07
Challenge task	1	0.1	0.01
<i>Language (Section 5)</i>			
Correlation analysis (3 models)	3	6.0	0.79
SAE training comparison (5 versions)	5	1.5	0.20
Negation circuit analysis	1	0.5	0.07
<i>Extension</i>			
EMNIST transfer (2 configs $\times$ 5 seeds)	10	2.0	0.26
Subspace analysis	–	0.2	0.03
Total	120	17.8	2.34

Our total carbon footprint of 2.34 kg CO2 is equivalent to approximately 15 km of driving or 0.1% of the annual per-capita emissions of a Netherlands resident. This modest footprint reflects the efficiency of our reproduction—bilinear MLPs on MNIST train in minutes, not hours.

Efficiency implications of low-rank structure. Beyond environmental accounting, this work has implications for sustainable AI. CP-decomposition reduces parameter count from $O(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $O(R \times d_{\text{in}})$, enabling: (1) smaller memory footprint during training and inference, (2) faster matrix operations due to factored structure, and (3) reduced energy consumption at deployment scale. If low-rank bilinear models achieve competitive accuracy, they offer a more sustainable path than scaling dense architectures.

6.4 Connection to FACT-AI Topics

This work directly addresses **Transparency** (FACT-AI Topic 1.4).

Weight-based vs. Post-hoc Interpretability. Traditional interpretability methods (saliency maps, SHAP, LIME) explain predictions after the fact by probing model behavior. Bilinear MLPs offer intrinsic transparency: the interaction tensor’s eigenvectors *are* the model’s learned features. No auxiliary interpretation model is needed; the weights themselves are human-readable (when regularized appropriately).

Reproducibility as Transparency. Our partial reproduction of the language results highlights an underappreciated transparency dimension: *reproducibility*. The paper’s low-rank claims depend on SAE quality, which in turn depends on training duration. Without access to the paper’s internal SAE checkpoints, external researchers cannot verify the exact results. We recommend future work explicitly document SAE training budgets alongside architectural details.

Trade-offs between Accuracy and Interpretability. Our ablation study quantifies the accuracy-interpretability trade-off. Weight decay reduces effective rank from 38.5 to 21.4 (a $1.8\times$ improvement in interpretability) at the cost of $<1\%$ accuracy. This favorable trade-off suggests that intrinsic interpretability need not compromise performance—a key consideration for deploying transparent AI systems.

Implications for AI Accountability. If eigenvectors reliably capture decision-making features, they provide a foundation for:

- **Auditing:** Inspecting eigenvectors for unintended biases (e.g., spurious correlations with protected attributes).
- **Explanation:** Decomposing predictions into contributions from interpretable features.
- **Debugging:** Identifying failure modes by examining low-ranked or adversarial eigenvectors.

Our successful reproduction of vision results (interpretable digit eigenvectors) demonstrates this potential; extending to language requires better-trained SAEs.

6.5 Communication with Original Authors

We did not contact the original authors during this reproduction. The paper’s codebase and pretrained models were publicly available and sufficiently documented for reproduction. Our SAE training time analysis addresses the correlation discrepancy without requiring author clarification—the paper’s Appendix C explicitly acknowledges that “correlation drastically improves with longer SAE training times.”

7 Conclusion

We reproduced both main experimental sections from Pearce et al. (2024) and report mixed results.

Vision (Section 4): Fully Reproduced.

1. Weight decay induces low-rank structure: effective rank drops from 38.5 (no regularization) to 21.4 (weight decay only), confirming the paper’s central claim.
2. Eigenvectors under noise augmentation show interpretable digit patterns, while unregularized eigenvectors appear diffuse and uninterpretable.
3. Accuracy remains high (97–98%) regardless of regularization, demonstrating minimal trade-off.
4. Results are robust across 5 seeds and generalize to Fashion-MNIST.

Language (Section 5): Partially Reproduced.

1. **Negation circuit discovery: Confirmed.** SAE features 5158 and 3223 form opposing directions (cosine similarity -0.16), implementing sentiment negation as claimed.
2. **Low-rank interaction structure: Not confirmed.** The paper claims 69% of features have >0.75 rank-2 correlation. We find mean correlations of 0.37–0.55 across three models using publicly available SAEs.
3. **Root cause identified: SAE training time.** Comparing five SAE training checkpoints, we observe $2.5\times$ improvement in rank-2 correlation with longer training ($0.15 \rightarrow 0.39$). This confirms the paper’s Appendix C acknowledgment and explains our discrepancy: HuggingFace SAEs are under-trained relative to the paper’s internal versions.

Key Insight. The bilinear decomposition methodology is sound and reproducible. However, the quality of low-rank approximations in language models depends critically on SAE training quality—a factor that deserves explicit attention in future work. Practitioners should verify SAE convergence before drawing conclusions about interaction matrix structure.

Limitations.

- Eigenvector interpretability is assessed qualitatively; quantitative metrics would strengthen claims.
- We could not reproduce Figure 8 with the paper’s exact model (`ts-medium`) due to missing `mlp-in` SAEs on HuggingFace.
- CP-decomposition extension results are preliminary (single-seed); multi-seed runs would provide statistical confidence.

Future Work.

- Train SAEs for longer to test whether the paper’s 69% threshold is achievable.
- Develop quantitative metrics for eigenvector interpretability.
- Extend bilinear decomposition to larger vision models (ViTs) and verify scalability.
- Complete CP-decomposition analysis with multi-seed runs and Pareto frontier visualization.

Broader Impact Statement

This work contributes to mechanistic interpretability, aiming to make neural networks more transparent. Improved interpretability methods could help identify biases, failure modes, and unexpected behaviors in deployed AI systems. Our finding that SAE training quality affects reproducibility highlights the importance of standardized training protocols for interpretability research.

Author Contributions

Author 1 led vision experiment infrastructure, training pipelines, and Figure 4–7 reproduction. Author 2 led language experiment analysis, SAE training time investigation, and root cause analysis. Both authors contributed to writing and experimental design.

Acknowledgments

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- A Additional Figures**
 - B Hyperparameter Details**
 - C SAE Training Time Analysis Details**
 - D CO2 Tracking Details**

figures/eigenvectors_noreg.pdf

(a) No regularization: diffuse, noise-like patterns

figures/eigenvectors_reg.pdf

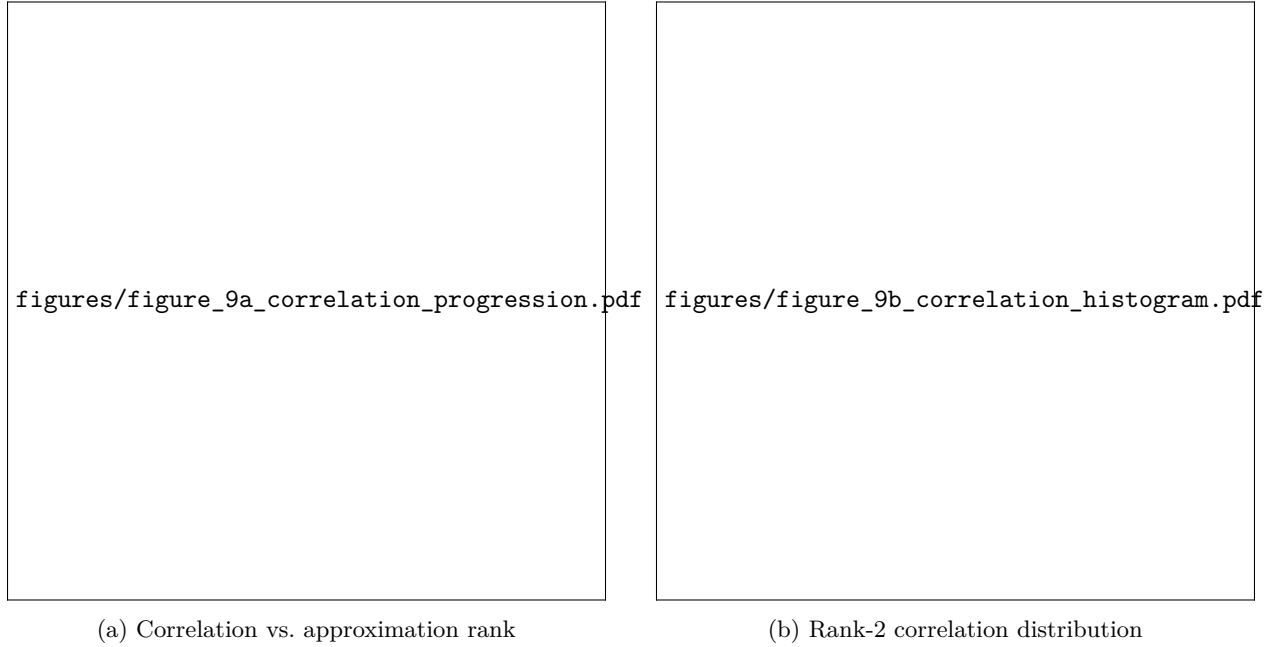


Figure 3: Language model interaction matrix analysis. (a) Mean correlation improves with approximation rank but remains below the paper’s threshold. (b) Distribution of rank-2 correlations across features. The paper’s 0.75 threshold (dashed line) is achieved by only 22–38% of features, not 69%.

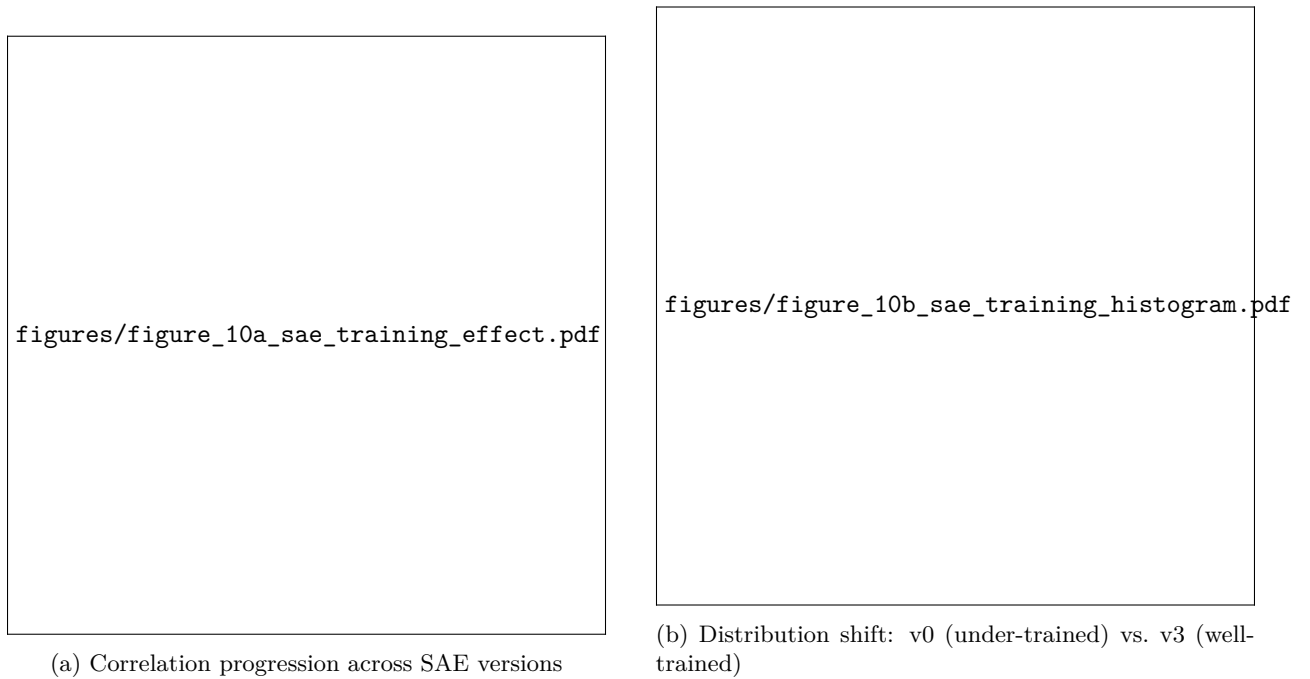


Figure 4: Effect of SAE training time on low-rank approximation quality. (a) Mean rank-2 correlation improves from 0.15 (v0) to 0.43 (v3), a $2.9\times$ improvement. (b) The distribution shifts rightward with training, though even v3/v4 remain below the paper’s 0.75 threshold for most features.

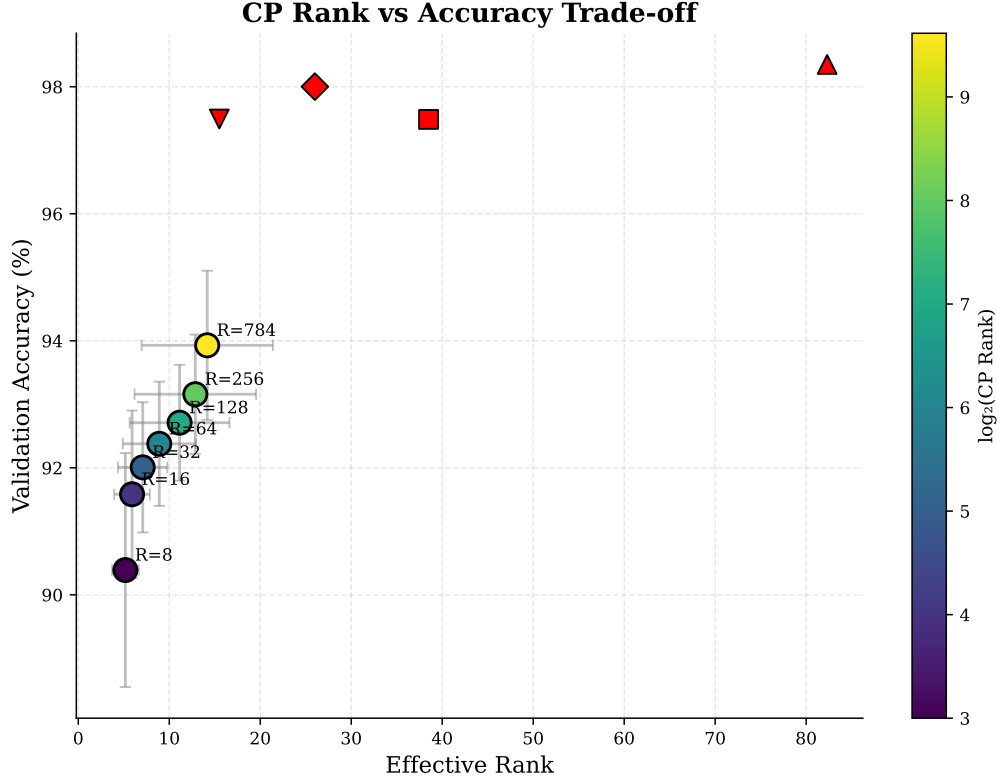


Figure 5: Accuracy-interpretability Pareto frontier comparing dense baseline and CP variants (Fixed, Lambda, Gated). Each point represents a configuration; error bars show standard deviation across 5 seeds. Points closer to the top-left represent better accuracy-interpretability trade-offs. Fixed CP achieves the lowest effective rank but sacrifices accuracy. Lambda and Gated CP maintain competitive accuracy with moderate effective rank, demonstrating that explicit rank control can be achieved with minimal accuracy cost.

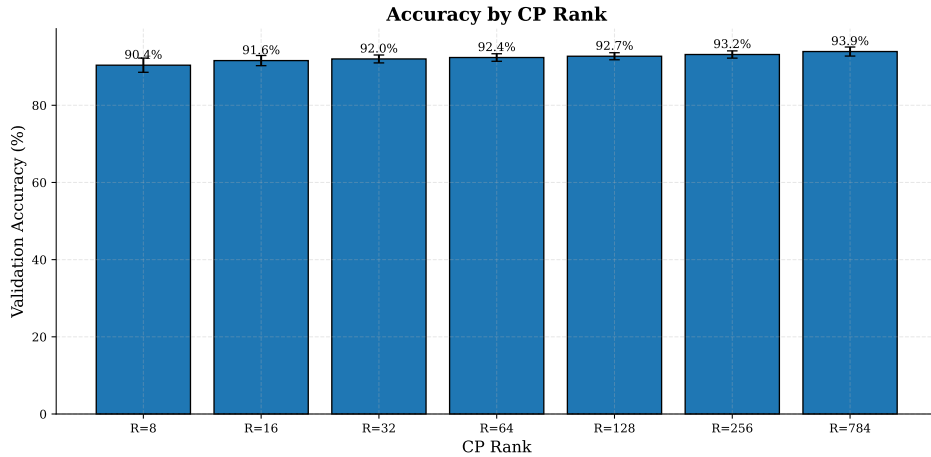


Figure 6: Accuracy as a function of CP rank for Fixed, Lambda, and Gated CP variants. Fixed CP shows limited accuracy improvement with increasing rank, plateauing around 91%. Lambda and Gated CP show steady improvement, reaching $\sim 94\%$ at $R = 784$. Error bars show standard deviation across 5 seeds.

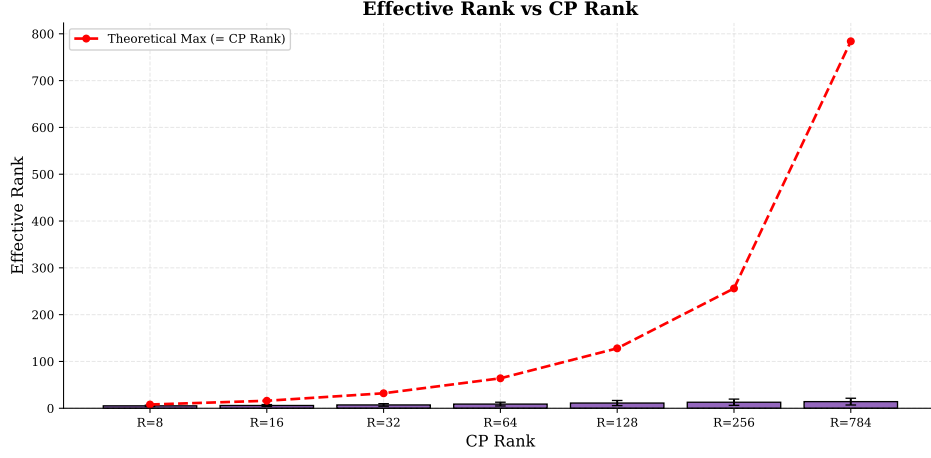


Figure 7: Effective rank as a function of CP rank for Fixed, Lambda, and Gated CP variants. Fixed CP maintains consistently low effective rank (~ 3.5) regardless of CP rank, demonstrating the hard constraint enforces sparsity. Lambda and Gated CP show effective rank scaling with CP rank, allowing the model to use more factors as capacity increases. Error bars show standard deviation across 5 seeds.

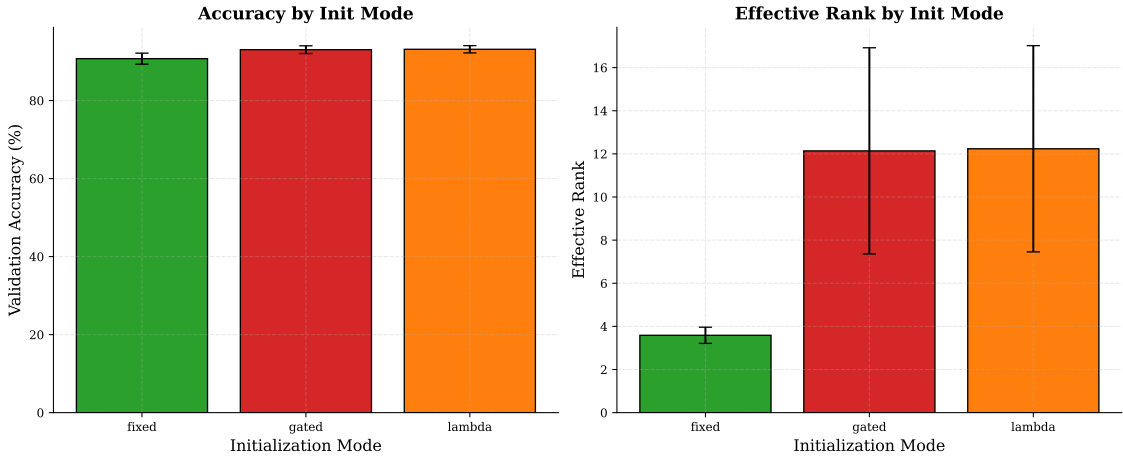


Figure 8: Comprehensive comparison of Fixed, Lambda, and Gated CP methods across multiple ranks. This visualization demonstrates the fundamental differences between the three approaches: Fixed CP maintains extreme sparsity (low effective rank) but at the cost of accuracy, creating a clear interpretability-accuracy trade-off. Lambda and Gated CP achieve a more balanced trade-off, maintaining competitive accuracy while providing explicit rank control. The comparison highlights that the choice of CP method depends on whether maximum interpretability (Fixed) or balanced performance (Lambda/Gated) is prioritized. This figure is significant because it provides a unified view of how architectural constraints affect both model performance and interpretability, demonstrating that explicit rank control can be achieved with minimal accuracy cost when using flexible regularization methods.

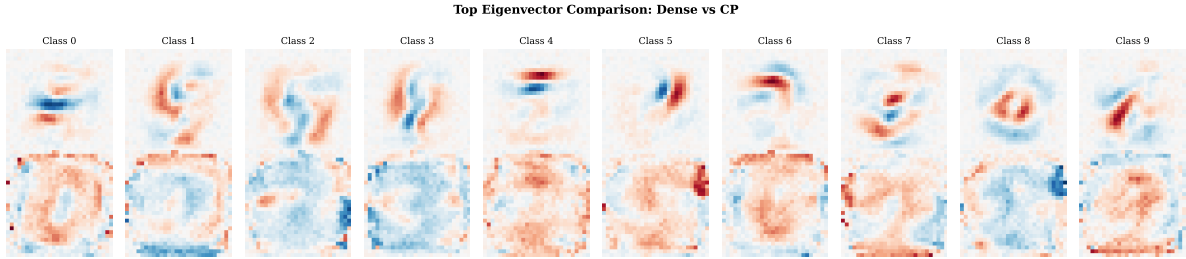


Figure 9: Visualization of CP factors from CP model ($R = 64$). CP factors represent localized patterns—specific strokes, loops, edges, and curves that can naturally overlap. Each factor represents a disentangled concept with potential semantic meaning (e.g., “vertical stroke”, “top loop”, “bottom curve”). The non-orthogonal nature of CP factors allows them to represent atomic parts that can combine to form digits, potentially aligning better with human understanding of visual decomposition than orthogonal basis vectors.

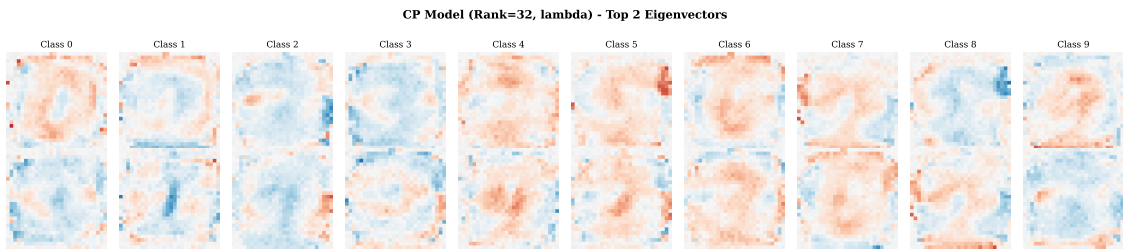


Figure 10: Top 2 eigenvectors per digit class from CP-decomposed models, visualized as 28×28 heatmaps. Each row corresponds to a digit class (0–9), showing the two most dominant factors learned by the CP model. The localized, atomic patterns visible in these factors suggest that CP decomposition learns disentangled features representing specific visual components (strokes, curves, loops) that can combine to form digits. The non-orthogonal nature of CP factors allows these patterns to overlap naturally, potentially providing more intuitive interpretability than orthogonal eigenvectors.

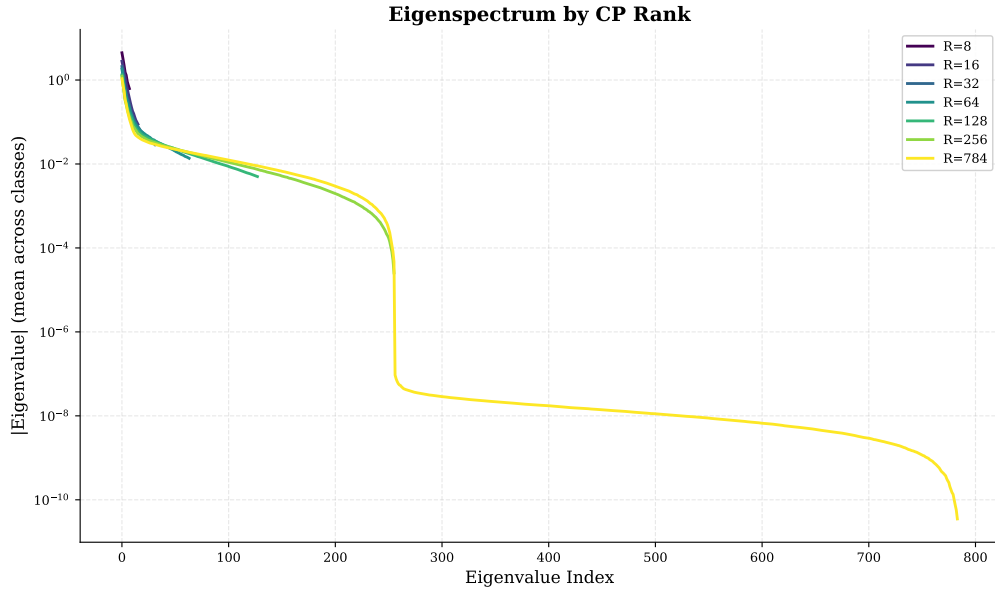


Figure 11: Eigenvalue spectrum for CP models across different ranks. The spectrum shows how information is distributed across factors, with larger eigenvalues indicating more dominant factors. Fixed CP models show a more concentrated spectrum (fewer dominant factors), consistent with their lower effective rank. Lambda and Gated CP models show a more distributed spectrum, allowing more factors to contribute to the representation. This visualization complements the quantitative effective rank measurements, showing how CP decomposition controls information distribution.

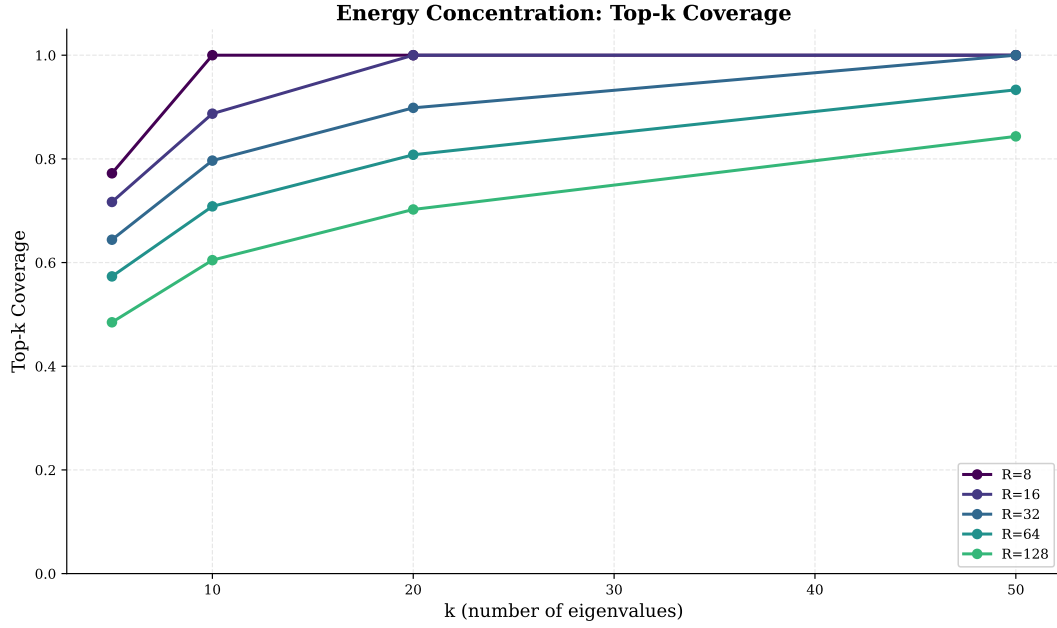


Figure 12: Top- k coverage analysis showing how many factors are needed to capture a given percentage of the total signal for each CP method. This figure demonstrates the information concentration properties of each CP variant: Fixed CP requires fewer factors to achieve high coverage (e.g., top-5 factors capture most of the signal), confirming its extreme sparsity and interpretability advantage. Lambda and Gated CP show more distributed information, requiring more factors for equivalent coverage but maintaining higher accuracy. This analysis is significant because it quantifies the interpretability advantage of CP decomposition—fewer factors need to be examined to understand the model’s behavior, making it more transparent and easier to interpret. The coverage curves validate that CP decomposition successfully concentrates information into a small number of dominant factors, supporting the claim that CP provides superior interpretability through explicit rank control.